

Electronic Circuit Design (EE2001)

Date: September 24th 2024

Course Instructor

Dr. Omer Saleem

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3



Roll No



Section



Student Signature

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Attempt all the questions.

CLO # 1: Analyze basic amplifier circuit models.

- Q1: A current amplifier for which $R_i = 1 \text{ k}\Omega$, $R_o = 10 \text{ k}\Omega$, and $A_{is} = 100 \text{ A/A}$ is to be connected with a 100.0 mA source with a resistance of $100 \text{ k}\Omega$ and a load of $1 \text{ k}\Omega$. What are the values of current gain i_o/i_s and voltage gain v_o/i_s , expressed in ratio and in dB. [10 marks]

CLO # 2: Design electronic circuits using op-amps.

- Q2: Design an op-amp circuit to implement the following mathematical operation.

$$V_o = 10V_1 + 5V_2 - 2V_3 - 4V_4$$

Use strictly $100 \text{ k}\Omega$ resistors in the feedback path of the designed circuit. Clearly draw the circuit and evaluate the resistances. [10 marks]

CLO # 2: Design electronic circuits using op-amps.

- Q3: Assuming the op-amp to be ideal, it is required to design the circuit shown in the following figure to implement a current amplifier with gain $i_L / i_1 = 21 \text{ A/A}$. [10 marks]

- What is the current in the $10 \text{ k}\Omega$ resistor?
- Find v_x in terms of i_1 and i_2 . Hence, express i_2 in term of i_1 .
- Find the required value of R .

